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课程名称：科技英语文献阅读与翻译

English Scientific Literature Reading and Translation (A)

Part I Scientific Literature Learning (20 points)

Directions: In this section, you will decide which is the best answer according to general knowledge of scientific literature. Then mark the corresponding letter on the Answer Sheet with a single line through the center.

- A** 1. Which of the following does **NOT** belong to scientific literature?

 - A) Novels and stories
 - B) Documents about achievements in science
 - C) Documents about developments in technology
 - D) Documents about researches in science and technology
- C** 2. Which of the following is **NOT** categorized as a product specification?

 - ☒ A) Financial product specification
 - ☒ B) Industrial product specification
 - C) Publishing product specification
 - ☒ D) Agricultural product specification
- C** 3. The following are published in academic journals **EXCEPT FOR** _____.

 - A) book reviews
 - B) review articles
 - C) science fictions
 - D) research articles
- D** 4. Which of the following is **NOT** the feature of technical reports?

 - A) Immediate ☒
 - B) Sequential
 - C) Confidential
 - D) Open to the public
- A** **C** 5. Which term below is **NOT** classified according to the range of their application?

 - A) Regional standards
 - B) International standards
 - C) Technical standards
 - D) National standards
- C** 6. How many years would a patent be protected?

 - A) 10
 - B) 30
 - C) 20
 - D) 40
- C** 7. What is the other name given to government publications?

 - A) Scientific literature
 - B) Agency publications
 - C) Official publications
 - D) Grey literature

D 8. A journal research article and a graduation paper are similar in the following aspects **EXCEPT FOR** _____.

- A) They both can be focusing on some unknown problems. ✓
- B) They both can be primary research. ✓
- C) They both can be written by university students. ✓
- D) They both can be written by more than one researcher.

A 9. Which of the following is **NOT** the function of citation indexes?

- A) Evaluating the financial performance of a given institute
- B) Retrieving the development of a given academic area
- C) Evaluating the academic performance of a given scholar
- D) Retrieving the latest literature of a given research problem

P 10. Which of the following is the document distributed after a conference?

- A) Invitations letters
- B) Daily schedules
- C) Opening speeches
- D) Conference reports

A 11. Which type of title does this title “Are there rearrangement hotspots in the human genome?” belong to?

- A) Interrogative title
- B) Declarative title
- C) Descriptive title
- D) Compound title

B 12. In the “author affiliation” section of a research article, the author marked with a litter star is called _____.

- A) the lead author
- B) the corresponding author
- C) the first author
- D) a co-author

D 13. Ethical issue will be involved if the research subjects are the following **EXCEPT FOR** _____.

- A) fetuses
- B) humans
- C) human tissues
- D) buildings

C 14. The result section of a research article usually presents the research findings in the form of _____.

- A) copies
- B) drawings
- C) tables and figures
- D) explanations

C 15. The conclusion section of a research article will contain the following information **EXCEPT FOR** _____.

- A) presentation of the future of the subject ✓
- B) restatement of the research question and thesis statement ✓
- C) description of how the research was carried out
- D) summary of the main points in the article ✓

16. In the acknowledgement section of a research paper, what was usually thanked **-EXCEPT FOR** _____.
- A) most important teachers ✓
 - B) leaders of your school
 - C) financial aids
 - D) financial support from a foundation
17. Which move is **NOT** included in the discussion section of a research article?
- A) Analyzing and discussing the results ✓
 - B) Answering the questions proposed in the study
 - C) Interpreting the results of research ✓
 - D) Discussing the rationale of doing the research
18. Which style does the in-text citation “Romantic poetry is characterized by the ‘spontaneous overflow of powerful feelings’ (Wordsworth 263)” belong to?
- A) MLA
 - B) APA
 - C) Vancouver
 - D) IEEE
19. When you read phrases in the introduction section of a research article, such as “little is, however, known about participants views of”, you will know that the following contents might possibly be about _____:
- A) what other researchers have already done ✗
 - B) the main theme of the paper ✗
 - C) what the researcher has finished ✗
 - D) preparing for the present research by pointing out the research gap
20. An in-text reference, also called the in-text citation, can be in the form of indirect citation which is ____.
- A) to paraphrase or summarize texts from original writing ✓
 - B) to explain what others said with no indication of citation
 - C) to paraphrase others’ words using quotation marks
 - D) to quote word by word of what others said

Part II Reading Comprehension (40 points)

Directions: There are 4 passages in this section. Each passage is followed by some questions or unfinished statements. For each of them there are four choices marked A), B), C) and D). You should decide on the best choice and mark the corresponding letter on the Answer Sheet with a single line through the center.

Passage One

Questions 21 to 25 are based on the following passage.

Who could have imagined that research on a little known bacterial plant pathogen (细菌性病原体), which ~~causes a disease called “crown gall,”~~ would revolutionize the future of agriculture? Yet that is what happened. In the early 1980s, research on the plant pathogen Agrobacterium (土壤杆菌) led to the first successful and dependable genetic engineering system for crops. Since then the science underpinning genetic engineering has come a long way. Genes from any species now can be specifically tailored to function not only in plants, but in certain plant tissues and at defined times. The genetic tools and databases

emerging from the rapidly developing field of genomics (基因组学) are making this even easier. Plant breeders now have a vastly expanded reservoir of possible genes for disease and pest resistance, drought tolerance, improved nutritional properties, and other beneficial traits that can be used to improve the profitability of farms, improve consumer health, and protect the environment. In short, the promise and potential of biotechnology is extraordinary. Yet as promising as agricultural biotechnology is, it has raised concerns on several fronts. For example, some critics contend the current state of knowledge is insufficient to be certain genetically engineered crops will not harm the environment or human health. Others believe wide-scale adoption of genetically engineered crops will shift control of genetic resources away from farmers to large multinational corporations thereby threatening food security for poorer countries.

This issue of Economic Perspectives explores many of the multifaceted policy questions associated with the worldwide debate on agricultural biotechnology. I hope this presentation will inform the reader and shed some light on a highly charged debate that often lacks a scientific foundation or an appreciation of what will be required of agriculture in the 21st century. You can be confident that the United States will continue to pursue agricultural biotechnology policies based on scientific analysis and a commitment to human health and safety.

21. Which one of the following is the main topic of this passage?

- B
- A) U.S. agricultural biotechnology policies. ✗
 - B) Biotechnology and its safety. ✓
 - C) Future agriculture.
 - D) Debate on genetic foods. ✗

22. Which statement is NOT true about Agrobacterium?

- C
- B
- A) The research on it had an immense influence on the future of agriculture. ✓
 - B) Agrobacterium was discovered as a result of the study of a disease called "crown gall." ✓
 - C) A great breakthrough was not made in the research on Agrobacterium until early 1980s. ✗
 - D) Agrobacterium is a kind of bacterial plant pathogen that is little known to the world before 1980s. ✓

23. Which one of the following is not the effect produced by the development of genetic engineering?

- B
- A) Genes can be applied to both plants and their tissues. ✓
 - B) All the people believe that genetic food is superior and can replace the status of the traditional food.
 - C) Genetic tools and databases appear as the result of the development of genomics. ✓
 - D) Genetic plants have many advantages over traditional plants. ✓

24. What does "promise" mean in "the promise and potential of biotechnology"?

- D
- A) A guarantee.
 - B) A declaration. ✗
 - C) A pledge.
 - D) Good prospects.

25. What's the author's argument over biotechnology?

- A) Genetically engineered crops might harm the environment or human health.
- B) Large multinational corporations will win the control of genetic resources from farmers if genetically engineered crops are widely adopted.
- C) Its disadvantages outweigh its advantages.
- D) Agricultural biotechnology policies should be based upon scientific analysis and a commitment to human health and safety.

Passage Two

Questions 26 to 30 are based on the following passage.

Television — that most widespread and persuasive of modern technologies, marked by rapid change and growth — is moving into a new era, an era of extraordinary sophistication and diversity (the quality of being diverse), which promises to reshape our lives and our world. It is an electronic revolution made possible by the marriage of television and computer technologies.

The word "television", derived from its Greek (tele: distant) and Latin (visio: sight) roots, can literally be interpreted as sight from a distance. *Very simply put*, it works in this way: through a sophisticated system of electronics, television provides the capability of converting an image (focused on a special photo-conductive plate within a camera) into electronic impulses, which can be sent through a wire or cable. These impulses, when fed into a receiver (television set), can then be electronically reorganized into that same image.

Television is more than just an electronics system, however. It is a means of expression, as well as a vehicle for communication, and as such becomes a powerful tool for reaching other human beings.

The field of television can be divided into two categories determined by its means of transmission. First, there is broadcast television, which reaches the masses through air-wave transmission of television signals. Second, there is non-broadcast television, which provides for the needs of individuals or specific interest groups through controlled transmission techniques.

Traditionally, television has been a medium of the masses. We are most familiar with broadcast television because it has been with us for about thirty-seven years in a form similar to that exists today. During those years, it has been controlled, for the most part, by the broadcast networks, ABC, NBC, and CBS, who have been the major providers of news, information, and entertainment. These giants of broadcasting have actually shaped not only television but our perception of it as well. We have come to look upon the picture tube as a source of entertainment, placing our role in this dynamic medium as the passive viewer.

26. What seems to be the best title for the passage?

→ revolution

- A) A Powerful Tool of Entertainment. ✓
- B) The Means of Transmission of TV. ✗
- C) TV: Its Working Principles and Functions. ✗
- D) A Brief Introduction to the Development of TV. ✓

27. The rapid change and growth of TV is due to the contributions of _____.

- A) computer technologies
- B) a new means of transmission
- C) a powerful vehicle of communication
- D) TV giants like ABC, NBC and CBS

最后一段！

28. What is the use of the second means of transmission?

- A) To meet the requirements of people especially interested in non-broadcast program. ✗
- B) To meet the requirements of certain groups of people.
- C) To meet the requirements of people with a special interest in cable TV. ✗
- D) To meet the requirements of people interested in controlled transmission techniques. ✗

29. The italicized "very simply put" in Para. 2 means "_____".

- A) when it is simply put down
- B) when it is put to simple tasks
- C) if it is put to simple tasks
- D) if it is expressed in a simple way

30. What is **NOT** the reason for TV to be powerful?

- A) It highlights broadcasting giants.
- B) It can communicate to human beings.
- C) It provides recreational programs.
- D) It provides news programs.

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Passage Three

Questions 31 to 35 are based on the following passage.

In science, a theory is a reasonable explanation of observed events that are related. A theory often involves an imaginary model that helps scientists picture the way an observed event could be produced. A good example of this is found in the kinetic molecular theory, in which gases are pictured as being made up of many small particles that are in constant motion.

A useful theory, in addition to explaining past observations, helps to predict events that have not as yet been observed. After a theory has been publicized, scientists design experiments to test the theory. If observations confirm the scientists' predictions, the theory is supported. If observations do not confirm the predictions, the scientists must search further. There may be a fault in the experiment, or the theory may have to be revised or rejected.

3 Science involves imagination and creative thinking as well as collecting information and performing experiments. Facts by themselves are not science. As the mathematician Jules Henri Poincare said: "Science is built with facts just as a house is built with bricks, but a collection of facts cannot be called science any more than a pile of bricks can be called a house."

Most scientists start an investigation by finding out what other scientists have learned about a particular problem. After known facts have been gathered, the scientist comes to the part of the investigation that requires considerable imagination. Possible solutions to the problem are formulated. These possible solutions are called hypotheses.

In a way, any hypothesis is a leap into the unknown. It extends the scientist's thinking beyond the known facts. The scientist plans experiments, performs calculations, and makes observations to test hypotheses. For without hypotheses, further investigation lacks purpose and direction. When hypotheses are conformed, they are incorporated into theories.

- B 31. According to the passage, a useful theory is one that helps scientists to _____.
A) find errors in past experiments ✗
B) make predictions
C) observe events ✗
D) publicize new findings ✗
- C 32. Bricks are mentioned in the third paragraph to indicate how _____.
A) mathematicians approach science
B) building a house is like performing experiments ✗
C) science is more than a collection of facts ✓
D) scientific experiments have led to improved technology ✗
- B 33. The author implies that imagination is most important to scientists when they _____.
A) evaluate previous work on a problem
B) formulate possible solutions to a problem
C) gather known facts
D) close an investigation
- C 34. What does the author imply is a major function of hypotheses?
A) Shifting through known facts. ✗
B) Communicating a scientist's thoughts to others. ✗
C) Providing direction for scientific research.
D) Linking together different theories. ✗
- D 35. Which of the following statements is supported by the passage?
A) Theories are simply imaginary models of past events.

- B) It is better to revise a hypothesis than to reject it. ✗
C) A scientist's most difficult task is testing hypotheses. ✗
D) A good scientist needs to be creative. ✓

Passage Four

Questions 36 to 40 are based on the following passage.

The great advantage of the electric train over the steam train, I think, is that it does not smell. When one is in a train drawn by a steam locomotive, one always has the unpleasant smell of burning coal around one. If one opens the window to look out at the view and to get some fresh air, one is covered with little bits of coal ash, which gets into one's hair and eyes, and soon forces one to shut the window again and withdraw into the impure atmosphere of one's carriage. In a tunnel one's misery reaches a peak, for there the smoke is forced into a small area, from which it escapes into the train itself through any small cracks it may find. In a long tunnel, the smell grows worse and worse, while one's eyes ache and the electric light in the carriage grows dim.

With an electric locomotive, on the other hand, one can enjoy the view and the fresh air without inconvenience. I really think that part of the extra joy one gets out of traveling through the Alps by train is due to the fact that the trains there are electrically operated, so that the traveler who has become used to having his air dirtied by smoke, and considers such a state of affairs as normal, believes that there is something extra special — indeed, almost magical — about the mountain air. Of course, mountain air is purer and fresher than the air of the plains, but the smoke of a steam train would not allow this state of affairs to last long.

- A 36. Why is one forced to shut the window again when one opens it in a steam train?
- A) Because bits of coal ash get into one's eyes and hair.
B) Because there is an unpleasant smell.
C) Because the unpleasant smell is around one.
D) Because the atmosphere in the carriage is impure.
- C 37. When does one feel most uncomfortable in a steam train?
- A) When the train reaches a peak ✗
B) When the smoke in a tunnel is forced into a small area.
C) When the train has entered a long tunnel.
D) When one's eyes ache and the electric light in the carriage grows dim. ✗
38. The writer enjoys traveling through the Alps by train because ____.
- A) the air is pure and fresh
B) there is nothing special about the air there ✗
C) the air there hasn't been dirtied by smoke yet ✓

D) the use of the electric train there prevents the air there from being dirtied

- C 39. What does the writer mean when he says that "the smoke of a steam train would not allow this state of affairs to last long?"
- A) The affairs of this country would soon be ruined by a steam train. ✗
- B) Affairs of this county would not last long because of the smoke of a steam train. ✗
- C) The fresh mountain air would soon be dirtied by a steam train. ✗
- D) The mountain air would not last long due to the smoke of a steam train. ✗
40. This passage is about _____. ✓
- A) the advantage of the electric train over the steam train ✓
- B) how fresh mountain air is dirtied ✗
- C) how to travel by train ✗
- D) a comparison between the electric and the steam train in all respects

Part III Sentence Translation (15 points)

Directions: Translate the following English sentences into Chinese. Please write your answers on the Answer Sheet.

1. More than three years have passed since Halley's Comet (哈雷彗星), our last naked-eye comet, graced the evening sky for Northern Hemisphere observers.
2. For 20 years we were passive witnesses to the ^{恶化} deterioration of prices of our raw materials and an excessive increase of the prices of manufactured goods.
3. The worldwide commercial plane market was valued at almost \$1.6 billion in 2019, and the sales could reach \$8.5 billion by 2027, according to ^{12!!!} Fortune Business. (财富)
4. The purpose of this book is to introduce you to these fundamental ideas, models, and results that permeate computer science, the basic concept of our field. 渗透
5. Since the product was introduced in 1997, it has not had an equal on the market, and three year's lead is a lot in the world.

Part IV Paragraph Translation (10 points)

Directions: Translate the following English paragraph into Chinese. Please write your answers on the Answer Sheet.

Astronomers (天文学家) no longer regard as fanciful (奇异的、稀奇的) the idea that they may one day pick up signals which have been sent by intelligent beings on other worlds. This possibility gives rise to interesting speculations (设想). Highly advanced civilizations may have existed on other planets long before intelligent forms of life evolved on the earth. Conversely, intelligent beings which are just beginning

to develop on remote worlds may be ready to pick up our signals in thousands of years' time, or when life on earth has become extinct. Such speculations no longer belong to the realm (范畴) of science fiction, for astronomers are now exploring the chances of communicating with living creatures (if they exist) on distant planets.

Part V Writing (15 points)

Directions: Please summarize the following reading passage into a short summary in English (100-150 words) according to your understanding of the content. Please write your answers on the Answer Sheet.

Most people can remember a phone number for up to thirty seconds. When this short amount of time elapses, however, the numbers are erased from the memory. How did the information get there in the first place? Information (that makes its way to the short-term memory (STM)) does so (via the sensory storage area). The brain has a filter which only allows stimuli (that is of immediate interest) to pass on to the STM, also known as the working memory.

There is much debate about the capacity and duration of the short-term memory. The most accepted theory comes from George A. Miller, a cognitive psychologist who suggested that humans can remember about seven chunks of information. A chunk is defined as a meaningful unit of information, such as a word or name rather than just a letter or number. Modern theorists suggest that one can increase the capacity of the short-term memory by chunking, or classifying similar information together. By organizing information, one can optimize the STM, and improve the chances of a memory being passed on to long-term storage. The capacity of STM can also be affected by diseases that cause neuro-degeneration, such as Alzheimer's disease which can be a factor in a person's short-term and eventually long-term memory. Damage to certain sections of the brain can disable the ability to think and recall memories stored in STM.

Long-term memory (LTM) is the final stage in which data can be stored for long periods of time. While short-term and working memory persists for only about 20 to 30 seconds, information can remain in long-term memory indefinitely. Long-term memory has also been called reference memory, because an individual must refer to the information in long-term memory when performing almost any task. Short-term memory is limited to a certain number of chunks of information, while long-term memory has a limitless store.

Understanding the information also makes it more retrievable. Retrieving information can be done by recognition or recall. Humans can easily recall memories that are stored in the long-term memory and used often, however, if a memory seems to be forgotten, it may eventually be retrieved by prompting. The more cues a person is given (such as pictures), the more likely a memory can be recalled. This is why multiple-choice tests are often used for subjects that require a lot of memorization.

Sample: Our brains save information in the short-term memory (STM) or the long-term memory (LTM). Our STM can be improved by organizing information and avoiding diseases. Recognition, recall and prompting are effective ways to regain information stored in the LTM. There are differences between the STM and LTM. While short-term memory lasts for a very short time, information can remain in long-term memory forever. While short-term memory is limited to a certain number of pieces of information, long-term memory has a limitless store.

Answer Sheet

Part III Sentence Translation (15 points)

得分	评卷人

1

2

3

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Part IV Paragraph Translation (10 points)

得分	评卷人

Keys

Astronomers (天文学家) no longer regard as fanciful (奇异的、稀奇的) the idea that they may one day pick up signals which have been sent by intelligent beings on other worlds. This possibility gives rise to interesting speculations (设想). Highly advanced civilizations may have existed on other planets long before intelligent forms of life evolved on the earth. Conversely, intelligent beings (which are just beginning

to develop on remote worlds) may be ready to pick up our signals in thousands of years' time, or when life on earth has become extinct. Such speculations no longer belong to the realm (范畴) of science fiction, for astronomers are now exploring the chances of communicating with living creatures (if they exist) on distant planets.

1. 三年多以前，北半球的天文观测者们在夜空中看到了哈雷彗星优雅的身影，它是我们能用肉眼看到的最后一颗彗星。
2. 我们在二十年里坐视原料价格暴跌和工业品价格暴涨。
3. 据《财富》杂志社报道，2019年全球商用飞机市场价值近16亿美元，到2027年销售额可能达到85亿美元。
4. 该书的目的是向读者介绍贯穿于计算机科学的基本思想、模型和结果，也就是该领域的基本概念。
5. 自从1997年产品推出以来，至今市场上还没有产品与之匹敌，在世界上领先三年实属不易。

Part IV

有朝一日会收到其他星球上智慧生物发出的信号，对于这种想法，天文学家不再认为荒诞无稽。这种可能性引发了有趣的设想。很可能远在地球智慧生物进化之前，其他行星上就已享有高度文明。反之，如果智慧生物刚刚开始遥远的星球孕育，或许要在数千年后，甚至地球上生命绝迹后，他们才可能有条件接收我们发去的信号。此种猜测不再属于科幻小说的范畴，因为天文学家正探索与遥远行星上的生物（如果它们存在的话）通讯的可能性。

Part V

Sample: Our brains save information in the short-term memory (STM) or the long-term memory (LTM). Our STM can be improved by organizing information and avoiding diseases. Recognition, recall and prompting are effective ways to regain information stored in the LTM. There are differences between the STM and LTM. While short-term memory lasts for a very short time, information can remain in long-term memory forever. While short-term memory is limited to a certain number of pieces of information, long-term memory has a limitless store.